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A fully integrated overshoot-reduction low-dropout regulator based on hybrid NMOS/PMOS power transistors technique for SoC applications

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ABSTRACT

This paper presents a fully integrated low dropout regulator (LDO) based on a hybrid NMOS/PMOS power transistors(HPT) technique. The HPT consists of a sub-power transistor NMOS and a main power transistor PMOS. Using the proposed HPT in the LDO, the high dc gain is achieved under heavy load condition and stability requirements are relaxed. With no extra power consumption, the proposed HPT provides a high charging rate at the gate of the main power transistor during a high-to-low load step. So the overshoot is reduced during load transient. Besides, the proposed LDO achieves high current efficiency by employing a current-recycling folded cascode amplifier. The post layout simulation results show that the LDO has a quiescent current of 7.4 μA and power supply rejection (PSR) less than −30 dB at 1 MHz. For large load transient of 100 μA to 20 mA and back to 100 μA with C_L = 100 pF and edge time of 1 ns, the overshoot and undershoot are 148 mV and 284 mV respectively. The proposed LDO is designed in SMIC 0.18 μm CMOS high-threshold-voltage process and occupies an area of 0.0252 mm^2.

1. Introduction

With the increasing number of data acquisition, data processing systems, and other systems integrated on a small scale, power management circuits are required to provide several different levels of stable voltages for the systems. In most applications, the supply voltage is unique and therefore results in different efficiencies for different voltage converters. While it is possible to increase the efficiency of power supplies with switch mode regulators, they are costly in terms of silicon area, and the magnetic elements are bulky and cause electromagnetic interference (EMI) [1]. For example, ripple and noise in the output voltage of switching regulators might disturb the normal operation of RF circuits. In contrast, the low dropout regulators (LDOs) can provide accurate and stable voltages for noise sensitive modules with a small chip area and it can support varying loads.

Fig. 1 shows the general flow of data acquisition and processing [2], where the digital signal processing module requires the LDO to achieve a fast transient response due to the presence of logic gates flipping on a large scale. In addition, LDOs are required to have an excellent voltage regulation under heavy load in order to protect converters from variations in input voltage during data acquisition. On the other hand, LDOs in the system may not all achieve high power efficiency. For example, when different modules require an input voltage of 2.5 V and 1.2 V respectively, and assuming that the power transistor have a minimum dropout voltage 200 mV, this may require a power supply with a minimum voltage of 2.7 V. The large dropout voltage, defined as the voltage difference between the unregulated supply voltage and regulated output voltage, allows for more options in LDOs structure.

Fig. 2 shows the schematic of conventional LDOs, which consists mainly of an error amplifier and a power transistor, where the power transistor of LDOs can be either a PMOS transistor or an NMOS transistor. LDOs with the PMOS power transistor(Fig. 2(a)) can achieve high power supply efficiency due to the low dropout voltage [3–5]. However, it generates large overshoot and undershoot when faced with rapidly changing loads [6] as its output impedance is higher than that of the NMOS power transistor, which results in poor dynamic response. Besides, it is not always possible to support a wide-range of load capacitance due to the stability requirements. During recent years, several techniques have been proposed to improve the dynamic response of LDOs [7–9] using the PMOS power transistor. In [7], the dynamic response of LDOs is improved by adding an additional transient enhancement circuit. A fast dynamic response is achieved in [6] by detecting spikes to adjust the charge and discharge rate of the gate capacitor. However, these techniques undoubtedly increase the power consumption of LDOs under no load condition. Compared to LDOs with the PMOS power transistor, LDOs with the NMOS power transistor(Fig. 2(b)) have low output impedance to reduce load-related stability problems and have better transient performance [10–14] as

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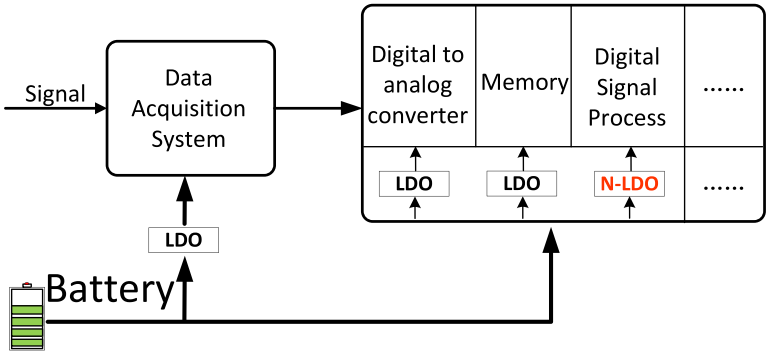


Fig. 1. A simple example of a battery-powered system in the data acquisition and data processing modules.

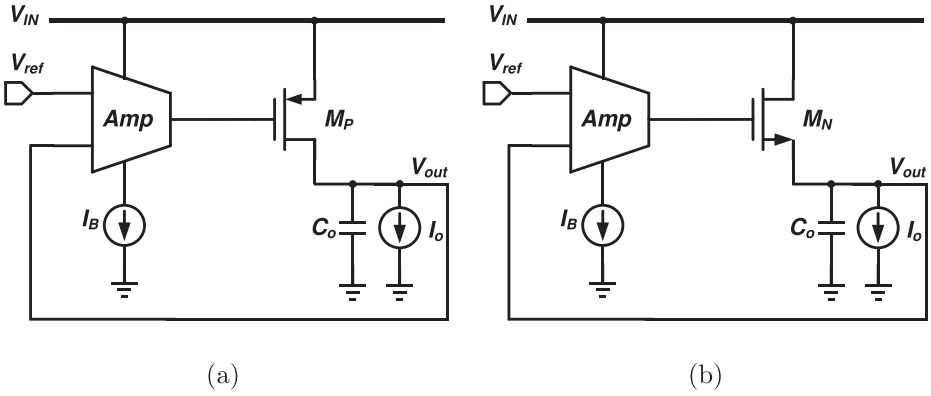


Fig. 2. (a) Typical structure of a PMOS-LDO, (b) Typical structure of an NMOS-LDO.

the signal returns directly from the source of power transistor to the gate of power transistor. However, they have a lower loop gain under heavy load condition since NMOS pass device acts as a source follower and its gain decreases as load current increases, which leads to a reduction in voltage and load regulation. Similarly, the adaptively biased technique suffers from the problem. In [11,15,16], an adaptively biased error amplifier boosts the bandwidth of the system and the rate of charging to the gate under heavy load condition resulting in reduced transient response time. But the amplifier output impedance decreases under heavy load condition, which leads to a reduction in amplifier gain and a reduction in the loop gain of the adaptively biased LDO. Consequently, these technologies may have some negative effects on the system.

Considering the above-mentioned advantages and disadvantages of NMOS and PMOS pass device and the limitations of previous technologies, this paper proposes an LDO with the hybrid power transistors(HPT) technique, which inherits the advantages of NMOS pass device, thus achieving better load transient and supporting wide-range of load capacitors. Moreover, it can maintain a relatively stable loop gain under all load current conditions. In addition, the proposed LDO consumes very little quiescent current and does not require complex frequency compensation. The LDO is designed in the 0.18 μm CMOS high-threshold-voltage process and the performance of the proposed structure is demonstrated by post layout simulations. The rest of the paper is organized as follows: the proposed HPT is discussed in Section 2. Section 3 describes the design of the proposed LDO. Section 4 gives the post layout simulation results. The conclusions are drawn in Section 5.

2. Proposed hybrid power transistors

A design of the HPT is proposed to improve the transient response of the LDO. Besides, it can improve the loop gain of the LDO under heavy load condition. The circuit structure is shown in Fig. 3. It contains NMOS, PMOS power transistor and current source, similar to the small and large power transistor structure [17]. The performance of the structure is effected by the amount of bias current. It can be noted that V_{out} remains constant and the current flowing through the NMOS power transistor is regulated by the amplifier due to the closed loop control. When I_L is less than I_{Set}, the bias current source enters the triode region and the PMOS power transistor turns off. When I_L is greater than I_{Set}, the current flowing through the NMOS power transistor is limited by the bias current source, and the PMOS power transistor is turned on with the current flowing through it as $I_L - I_{Set}$. Consequently, the low frequency gain of the HPT at different load conditions can be given by

$$A_{Light} = \frac{g_{mn} R_{L1}}{1 + g_{mn} R_{L1}} \tag{1}$$

$$A_{Heavy} = \frac{g_{mn} g_{mp} R_{L2} \left(r_{oN} \parallel r_{oB}\right)}{1 + g_{mn} R_{L2}} \tag{2}$$

$$g_{mn} = \sqrt{2\mu_n C_{OX} \left(\frac{W}{L}\right)_N I_{Set}} \tag{3}$$

where gm_n is the transconductance of M_n under light load condition, r_{oB} is the channel resistance of the bias current source, g_{mn} and g_{mp} are the transconductance of M_n and M_p under heavy load condition, respectively. It is obvious that the gain provided by the HPT structure in the LDO loop depends on the load. For the LDO to maintain a high loop gain under all load conditions, the value of I_{Set} needs to be determined. In this design, the maximum output current and output voltage of the LDO is 20 mA and 1.2 V respectively. Accordingly, the maximum load is $R_L = 60\ \Omega$. A relatively high g_{mn} can be obtained by increasing $(W/L)_N$ and I_{Set}. However, an excessively large I_{Set} causes $r_{oN} \parallel r_{oB}$ to decrease, which results in reduction of gain. Therefore, there is a tradeoff of the I_{Set} design and it is set to 1 mA in this design.

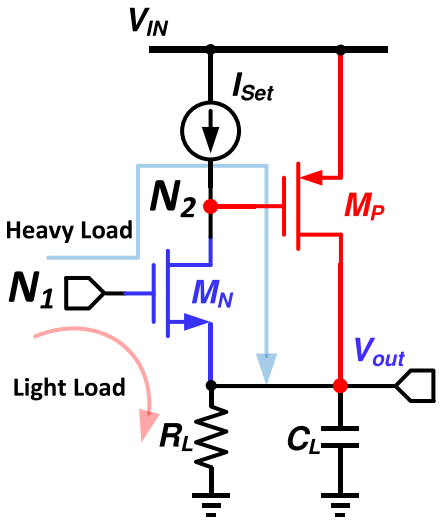


Fig. 3. The proposed HPT structure.

It should be noted that when the power transistor M_p operates properly, the output has a lower output resistance because of negative feedback. From a qualitative standpoint, when the voltage at N_1 is constant and V_{out} increases, the voltage increase at N_2. As a result, the current at the output of M_p decreases, and thus V_{out} decreases. The resistance looking in from the output can be given by

$$r_{out} = \frac{1}{g_{mn}g_{mp}\left(r_{oN} \parallel r_{oP}\right)} \tag{4}$$

It is clear from (4) that the open-loop output resistance of the LDO is reduced by a factor of $gm_p(r_{oN} \parallel r_{oB})$ under heavy load condition, which is not too large for the stability requirements.

The proposed HPT also improves the slewing of the gate drive at N_2 during load transient. During the step from heavy load to light load, the voltage at N_1 is regulated by the amplifier so that it decreases the output current of M_n, while the current source, which has not yet entered the triode region, generates a large transient current to charge the gate of M_p. As a result, the voltage at N_2 rises rapidly.

In short, the proposed HPT structure maintains a high loop gain without introducing additional poles in the bandwidth, which improves the load regulation of the LDO under heavy load condition. In particular, a large charging rate can be achieved for the M_p gate without consuming any additional quiescent current. In addition, the low output impedance not only simplifies the frequency compensation of the LDO but also enables stability over a wide-range of load condition, which will be further clarified in the next section.

3. Design of the proposed LDO

The proposed LDO is shown in Fig. 4. It employs a single-stage, current-recycling folded cascode amplifier, which consists of transistors $M_1 - M_{10}$ to maintain a high loop gain. Because the NMOS pass device does not provide additional gain, the loop gain essentially contributed by the error amplifier. As discussed in the previous section, the equivalent circuit structure of HPT at different loads is different. Under light load condition, M_p and M_{11} can be neglected and M_n realizes the source follower as the second gain stage of the LDO. Accordingly, the LDO can be regarded as a low output impedance two-stage amplifier driving a small on-chip capacitor, which is usually the effective input capacitor of the next stage of the on-chip system. Considering the stability of the LDO, the time constant at N_1 is much greater than that of N_3, therefore N_1 is used as the dominant pole of the system and N_3 as the non-dominant pole. The proposed LDO can be stabilized by compensating with a larger capacitor C_c, which can couple the parasitic capacitance of the M_n gate and thus achieves pole splitting under the full range of the load current. Similarly, under heavy load condition, the proposed LDO can also be treated as a two-stage amplifier with a feed-forward path. Because the pole at N_1 located outside of the unity gain frequency(UGF), it does not affect the stability of the proposed LDO. The details of the stability are discussed in the following subsection.

3.1. Stability analysis

The stability of the proposed LDO is studied using the loop-gain transfer function $H(s)$ of the regulation loop. Fig. 5 is the small-signal block diagram representation of the circuit in Fig. 4 for analyzing the loop-gain transfer function. Here, g_{mEA}, g_{mn} and g_{mp} represent the effective transconductance of the error amplifier, pass device M_n and pass device M_p, respectively. In addition, as discussed in the previous section, the R_{oeq} is approximately $1/g_{mn}$ under light load condition since the load resistance R_L is much greater than $1/g_{mn}$. The loop-gain transfer function is derived based on the following considerations.

(1) The source follower M_n has an approximate gain of 1 under light load condition. i.e. $R_{oeq}*(1/g_{mn}) \approx 1$. Under heavy load condition, $R_{oeq}*(1/g_{mn}) < 1$ as the R_L decreases.

(2) C_2, C_c and C_{gs} are of the same magnitude.

Based on the above, the loop-gain transfer function of the proposed LDO is given as

$$H(s)|_{0<I_L<I_{Set}} \approx \frac{-g_{mEA}R_1\left(1+\frac{C_{gs}}{g_{mn}}s\right)}{\left(1+R_1C_cs\right)\left(1+\frac{C_cC_{gs}+C_cC_L+C_LC_{gs}}{g_{mn}C_c}s\right)} \tag{5}$$

$$H(s)|_{I_L>I_{Set}} \approx \frac{-g_{mEA}g_{mp}g_{mn}R_1R_2R_{oeq}\left(1+\frac{C_2}{g_{mp}}s-\frac{C_2C_{gs}}{g_{mp}g_{mn}}s^2\right)}{\left(1+R_1C_cs\right)\left(1+\left(R_2C_2+\frac{C_c+C_{gs}}{C_c}R_{oeq}C_L\right)s\right)\left(1+\frac{R_2R_{oeq}C_2C_cC_L}{R_2C_2C_c+R_{oeq}C_L(C_{gs}+C_L)}s\right)} \tag{6}$$

From (5), there are two left-half-plane (LHP) poles, one LHP zero given by $P_1 = 1/R_1C_c$, $P_2 = g_{mn}/\left(C_{gs}+C_L+\left(C_LC_{gs}\right)/C_c\right)$, $Z_1 = g_{mn}/C_{gs}$. It can be observed that $UGF = g_{mEA}/C_c$. As explained in previous prerequisites, C_c and C_{gs} are of the same magnitude, and C_L typically is 50–100 pF as an on-chip capacitor. To ensure stability under light load, a minimum output current is required to make $P_2 > 2.2UGF$. As the load current begins to increase, g_{mn} also begins to increase accordingly. Therefore, the frequency of P_2 increases gradually and becomes much larger than UGF, which implies sufficient phase margin for light load condition. In addition, it should be noted that Z_1 is an LHP zero, however, it is placed at a higher frequency than the unity gain frequency and therefore does not improve the phase margin.

As the load current increases above I_{Set}, the open-loop transfer function of the LDO can be approximated as in Eq. (6). In fact, due to the combined effect of the decrease in source follower gain and the power stage gain from M_p, it can be approximated that the loop gain remains constant. Accordingly, the UGF has remained approximately unchanged. A second-order function of zeros implies that there are two zeros. Since the s^2 term is negative and the term s is positive [18], the LHP zero can be expressed as

$$z_1 = \frac{g_{mn}}{2C_{gs}}\left(\sqrt{1+\frac{4g_{mp}C_{gs}}{g_{mn}C_2}}-1\right) \tag{7}$$

the RHP zero can be expressed as

$$z_2 = -\frac{g_{mn}}{2C_{gs}}\left(\sqrt{1+\frac{4g_{mp}C_{gs}}{g_{mn}C_2}}+1\right) \tag{8}$$

Under heavy load condition, since $g_{mp} > g_{mn} \gg g_{mEA}$ and $C_c > C_{gs} > C_2$, $UGF \ll Z_1 < Z_2$ can be derived. Therefore, the RHP zero does not degrade the phase margin. In the two non-dominant poles, a reasonable estimate of the value of the non-dominant pole is necessary. According to the method of numerical analysis, the frequency of the non-dominant

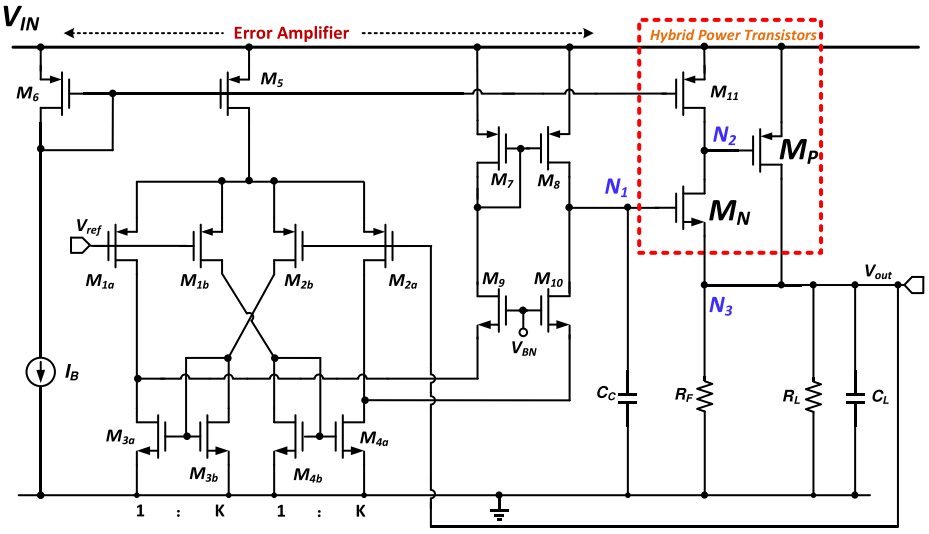


Fig. 4. Schematic of the proposed LDO design.

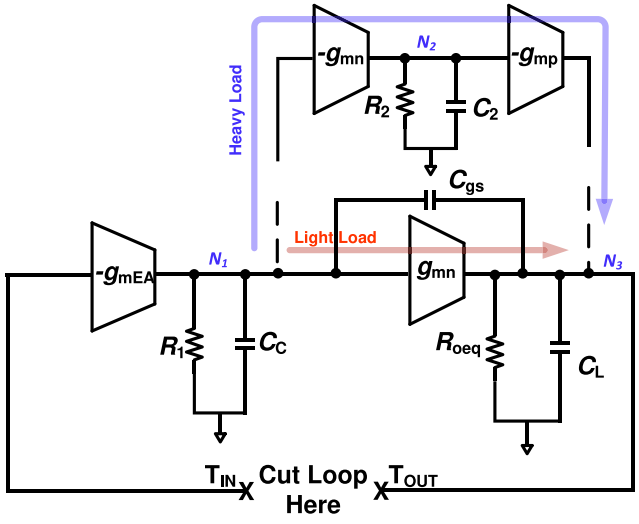


Fig. 5. Small-signal block diagram of the proposed LDO.

pole is much larger than the UGF. Thus, the theoretical phase margin is 90°. However, parasitic poles and zeros will degrade the phase margin.

The simulated loop-gain transfer function of the proposed LDO with C_L = 100 pF under different load condition are shown in Fig. 6. Simulated results validate the existence of only a single pole within the unity-gain frequency and a good phase margin of approximately 90° is obtained except for near no load condition. In addition, it is easily observed that the proposed LDO can maintain a stable loop gain and UGF at all load current. This simulation result further demonstrates that the frequency of the non-dominant pole is far greater than the UGF, thus ensuring the stability of the proposed LDO with the on-chip capacitor under different load conditions.

3.2. Load transient analysis

In capless-LDO, past research has given conventional LDOs their overshoot/undershoot formula with load current step, which can be approximately given by

$$\Delta V_o \approx (\Delta I_o / C_o)\left(\frac{1}{2\pi\omega_{\mathrm{UGF}}} + \frac{C_g \Delta V_g}{I_{\mathrm{SLEW}}}\right) \tag{9}$$

However, this does not apply to the proposed LDO with a HPT structure. When a step from I_{Heavy} to I_{Light} occurs, both NMOS pass device and PMOS pass device regulates the output current. Therefore, the above formula can be modified by

$$\Delta V_o \approx \left(\frac{I_{\mathrm{set}}}{C_o}\right)\left(\frac{1}{2\pi\omega_{\mathrm{UGF}}} + \frac{C_{g,n}\Delta V_{g,n}}{I_{\mathrm{SLEW,EA}}}\right) + \left(\frac{\Delta I_0 - I_{\mathrm{set}}}{C_o}\right) \times \left(\frac{1}{2\pi\omega_{\mathrm{UGF}}} + \frac{C_{g,p}\Delta V_{g,p}}{I_{\mathrm{SLEW,C}}}\right) \tag{10}$$

Here, ω_{UGF} is the loop bandwidth of the regulation loop and ΔV_g is the dc voltage variation at the gate of power transistors having gate capacitance of C_g to support ΔI_o change with a slewing current of I_{SLEW}. It should be noted that $I_{SLEW} < I_{Set}$ during the load current change, as the NMOS power transistor is not turned off during the regulation. On the other hand, the charging current at the gate of M_p is much greater than that of the error amplifier. It thanks to the fact that when output current of the NMOS power transistor is reduced, the current from the current source I_{Set} is not suddenly reduced, which allows to get a larger charging current at gate of PMOS power transistor. The simulated transient current at the gate of M_p and M_n is shown in Fig. 7 with the load current step from 20 mA to 100 μA in 1 ns. It can be observed that the charging current at the gate of M_p is even greater than the discharging current at the gate of M_n, which is further demonstrated that the HPT structure has a large slewing at gate of main power transistor.

3.3. Circuit design

In the schematic shown in Fig. 4, To ensure the high current efficiency and stability of the amplifier, a value of 2–3 is better for K, which always gives a gain greater than 60 dB under the full range of the load current. M_{11} is the current source, which mirrors the current of M_5. The accuracy of I_{Set} has little effect on the performance of the HPT structure and therefore the gate length of M_{11} can be reduced in order to reduce the chip area. However, the short gate length provides a small gain under heavy load condition. Therefore, there is a trade-off between area and gain. In addition, there is an on-chip capacitor C_c with a value of 12 pF, which is relatively large but ensures stability under the full range of the load current.

It must be noted that as part of the HPT, the NMOS sub-power transistor consumes a large voltage margin, which is characteristic of LDOs using the NMOS pass device. When $I_L > I_{Set}$, M_n is fully turned on and $V_{GS,N}$ of the NMOS power transistor reaches a maximum and remains constant. Consequently, the minimum supply voltage is

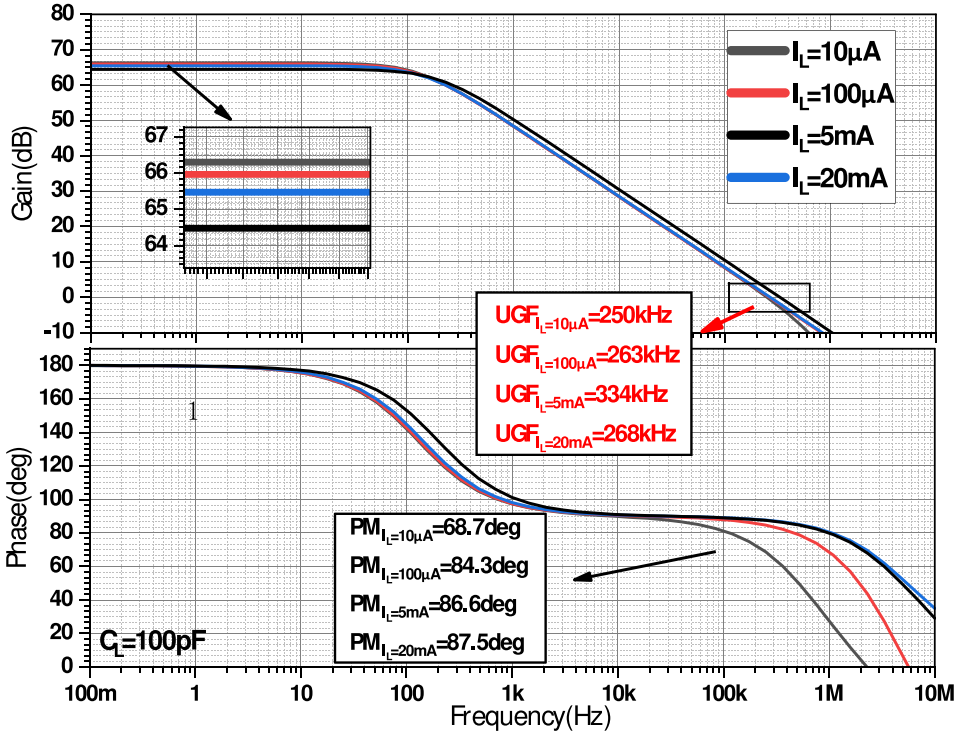


Fig. 6. The simulated loop-gain transfer function of the proposed LDO with C_L = 100 pF at I_L= 20 mA, 100 μA and 10 μA.

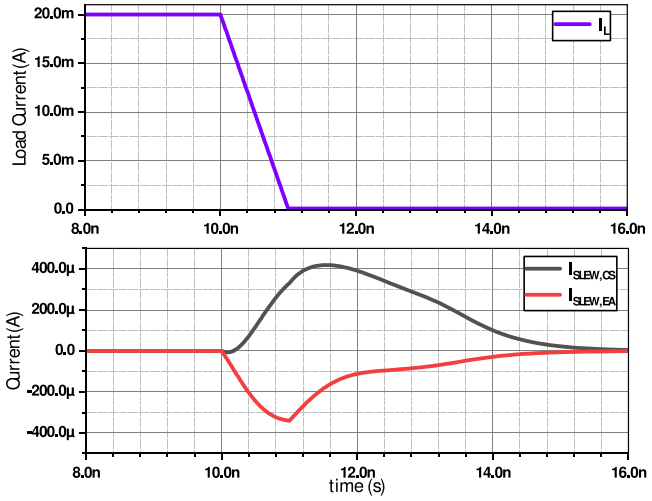


Fig. 7. The simulated currents at M_p and M_n gate with load current step from 20 mA to 100 μA in 1 ns.

V_{out}+$V_{GS,N}$+$V_{DS,8}$ to allow the error amplifier to maintain relatively high gain.

4. Simulation results and discussions

In order to verify the proposed HPT technology, the LDO in Fig. 4 has been designed and implemented using 0.18 μm CMOS high-threshold-voltage process. The transistor aspect ratios as well as feedback resistor and compensation capacitor values are reported in Table 1. Fig. 8 shows the layout of the proposed LDO. The effective area of this LDO is 0.0252 mm^2. The supply voltage is 3 V, the output voltage is 1.2 V, and the quiescent current is 7.4 μA.

The curves of the current provided to the output by M_N and M_P at different load currents are shown in Fig. 9. When the load current increases above 961 μA, the current supplied by M_N to the output no longer increases and M_P supplies current to the output.

Table 1
The proposed LDO component values.

Component	Values	Component	Values	Component	Values
$M_{1a,b}$	8/2	$M_{2a,b}$	8/2	M_{3a}	6/2
M_{3b}	3/2	M_{4a}	6/2	M_{4b}	3/2
M_5	4/1	M_6	2/1	$M_{7,8}$	8/2
$M_{9,10}$	20/4	M_{11}	100/0.5	M_N	1000/1
M_P	1000/0.3	C_c	12 pF	R_F	500 kΩ

Fig. 10 shows Phase margin(PM) under different combinations of load capacitance and load current conditions. The worst phase margin obtained is 9.1° for the load capacitance of 2 nF and load current of 1 μA, but PM > 45° in the region consisting of the points(1 μA, 65 pF), (157 μA, 2 nF), (20 mA, 2 pF), (20 mA, 1 pF) and (1 μA, 1 pF) and the remaining areas that do not satisfy the phase margins should be avoided.

The simulated line regulation of proposed LDO is shown in Fig. 11 while changing the supply voltage form 2.8 V to 3.6 V at 100 μA and 20 mA. The total error of the output voltage due to supply power variations is just 0.25%. This is caused by the decrease in gain of amplifier due to the reduction in supply voltage.

The simulated load regulation of proposed LDO is shown in Fig. 12. As proposed LDO is always able to maintain a high and essentially constant loop gain under different load conditions, its load regulation varies approximately linearly.

The load transient response is provided in Fig. 13. When load current changes from 100 μA to 20 mA, or back, with edge time of 1 ns at C_L = 100 pF, the LDO using only NMOS pass device has an overshoot and undershoot of 306 mV and 340 mV, respectively. After replacing HPT, the overshoot is reduced to 148 mV significantly. Due to the quiescent current of amplifier is only 7.4 μA, which results in slow load transient response. As a consequence, the slow recovery times can be observed at the output, which are 841 ns and 734 ns, respectively.

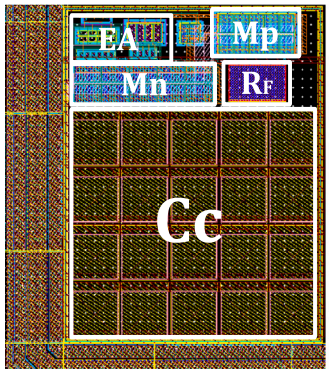


Fig. 8. Complete layout of the proposed LDO.

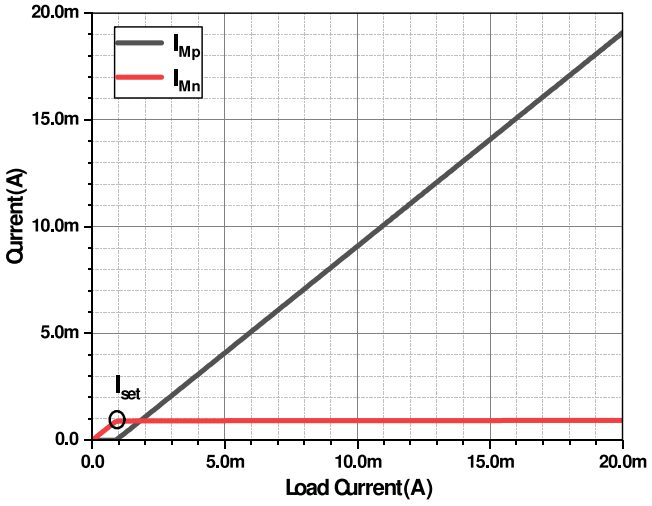


Fig. 9. Load current versus two power transistor currents.

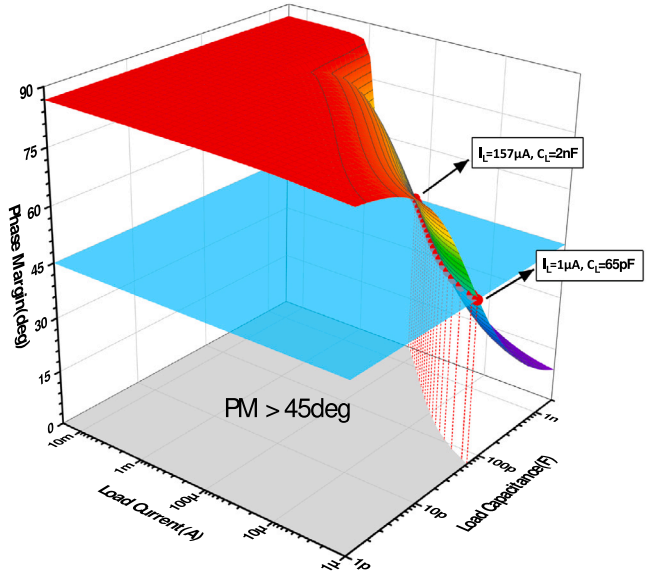


Fig. 10. Phase margin under different combinations of load capacitance and load current conditions.

Fig. 14 shows the load transient simulations of the proposed LDO for different load capacitances. The simulation conditions do not change

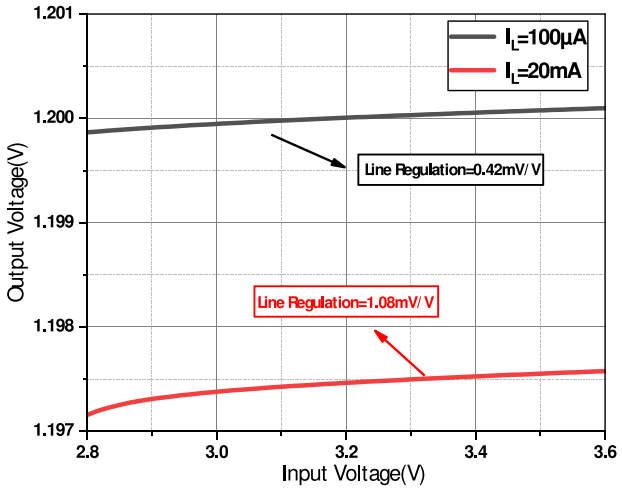


Fig. 11. The line regulation of the proposed LDO at I_L = 100 μA and 20 mA.

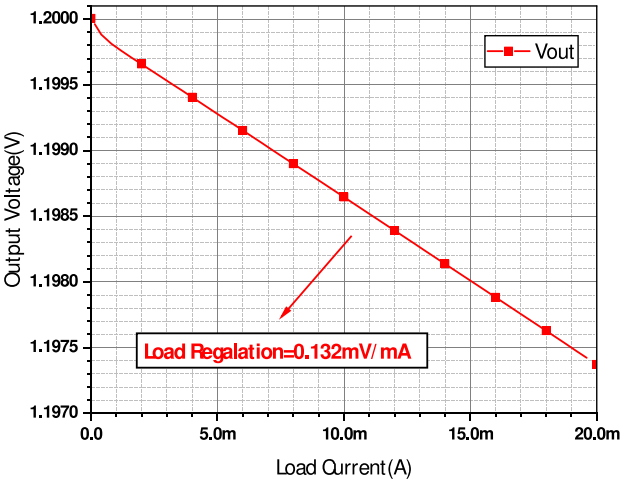


Fig. 12. Simulated load regulation of the proposed LDO.

except for the load capacitor. The worst settling time is 6.7 μs for the load capacitance of 2 nF and the worst overshoot is 148 mV for the load capacitance of 100 pF. Fig. 15 depicts the simulated transient response for different edge times with minimum and maximum load capacitance condition. It can be observed from Fig. 15 that there is a slight decrease in both overshoot and undershoot as the edge time increases.

The proposed LDO provides good performance on supply rejection ratio(PSR). As shown in Fig. 16, the LDO has at least 30 dB rejection ability at 10 MHz both under light and heavy load condition. The good PSR performance is achieved because the output resistance of the amplifier and the compensation capacitor C_c form a low-pass filter at the gate of NMOS power transistor.

Different performance parameters of the proposed LDO are summarized and compared with other capless-LDOs designed with different specifications using different technologies in Table 2. To compare their combined performance, the approach to calculating figures of merit(FOM) proposed in [19] is adopted, and the smaller FOM indicates the better performance. It can be expressed as

$$\mathrm{FOM} = T_R \times \frac{I_{qmin} + I_{qmax}}{2K\Delta I_{omax}} \tag{11}$$

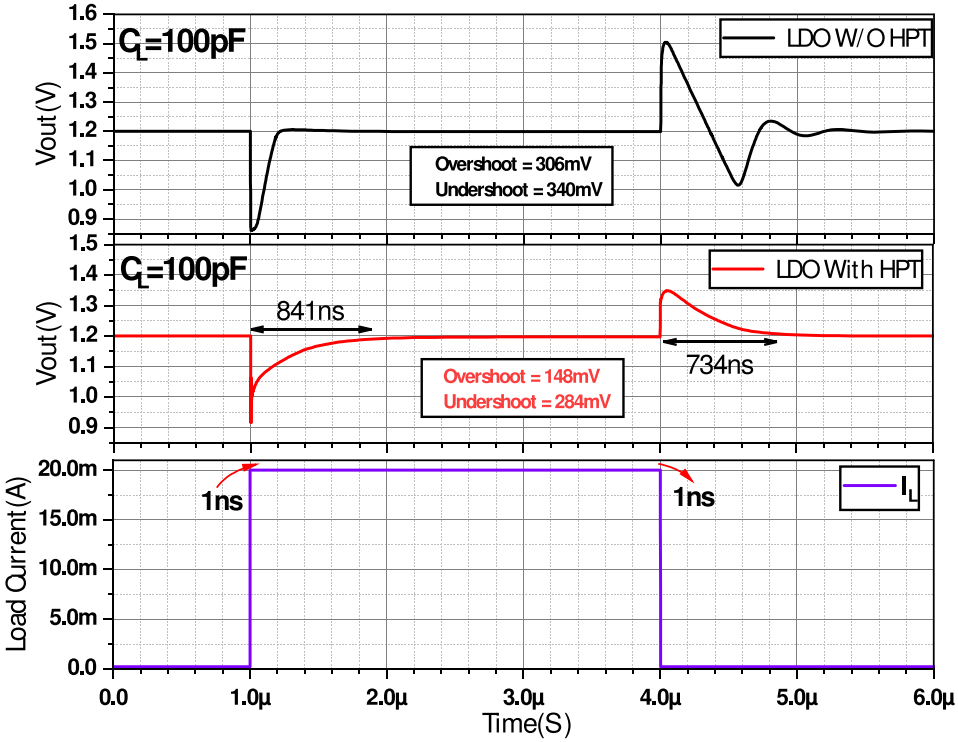


Fig. 13. Simulated load transient response with C_L = 100 pF.

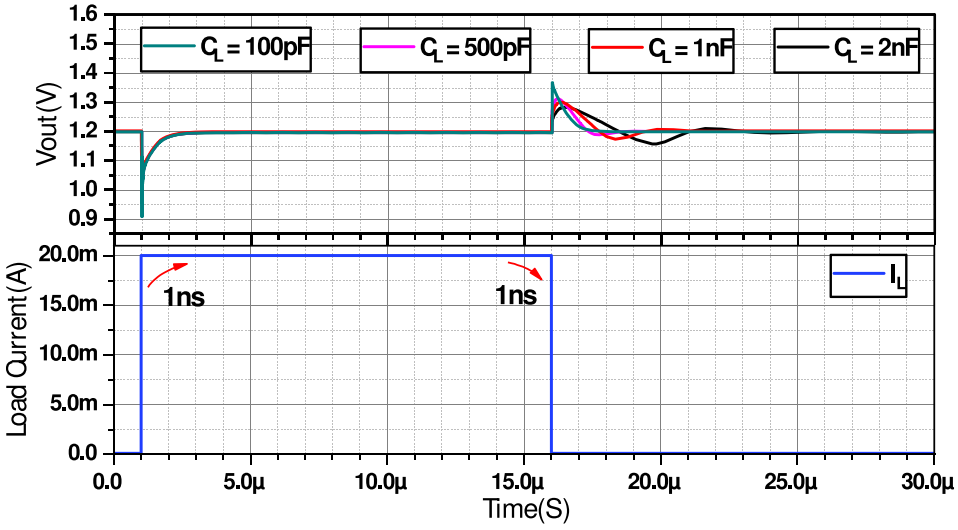


Fig. 14. Load transient simulations for the different load capacitances.

where K is a measure of the ratio between the different technologies, I_{qmin} represents the minimum quiescent current, I_{qmax} is the maximum quiescent current and I_{omax} represents the maximum load current [16]. T_R represents the transient response time, which is calculated by an approach based on the output voltage variation and the load transient measurement edge time T_{edge} and it should be expressed as

$$T_R = \begin{cases} \sqrt{\frac{2C_L \Delta V_{\text{out}}\ T_{\text{edgc}}}{\Delta I_{\text{omax}}}}, & T_R \le T_{\text{edge}} \\ C_L \frac{\Delta V_{\text{out}}}{\Delta I_{\text{omax}}} + \frac{T_{\text{edge}}}{2}, & T_R > T_{\text{edge}} \end{cases} \tag{12}$$

The proposed LDO can offer many advantages in some high voltage applications. Most of the previous work used a higher quiescent current to boost amplifier performance, increase gate slew rate or maintain stability at no load condition, whereas in this study, only I_q = 7.4 μA is required to achieve good performance. Although the works [13,20] have better load regulation than this study, the load regulation in this study remains essentially constant under all load conditions and consumes very low quiescent current. It is difficult to compare load transient with other studies under the same conditions. As far as the FOM is concerned, this study has a reasonably low value of FOM.

5. Conclusion

In this study, the HPT technique is proposed which proved effective in reducing overshoot during transient. The HPT provides a lower output impedance through negative feedback than conventional NMOS pass device and a large slew rate for gate of the main power transistor. The loop gain of the LDO is only reduced by 0.6 dB under heavy load

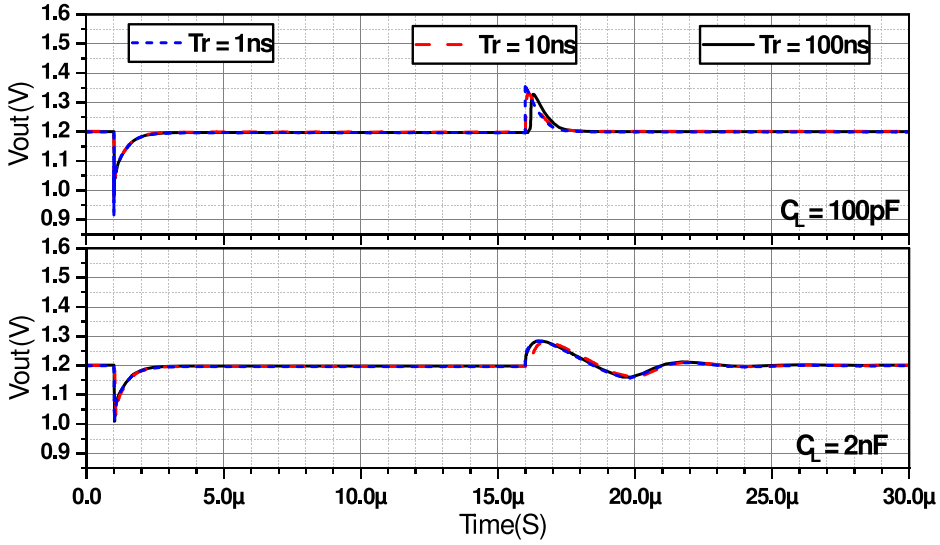


Fig. 15. The load transient response for different edge times with minimum and maximum load capacitance condition.

Table 2
Performance comparison with previously reported LDOs.

Parameter	**This work**[a]	[21][b]	[11][b]	[22][a]	[13][a]	[20][a]
Publication	**2021**	2020	2019	2018	2018	2017
Power type	**Hybrid**	PMOS	NMOS	PMOS	**Hybrid**	PMOS
Process (nm)	**180**	65	180	130	130	65
V_{in} (V)	**2.7–3.3**	1.05–1.2	1.6–1.8	1.4	1.3	1.5–2.5
V_{out} (V)	**1.2**	0.9	1.4–1.6	1.2	1.1	1.2
Load capacitance (pF)	**100–2000**	100	50	100	0–50	10
I_q (μA)	**7.4**	65	133	100	20.5–95	–
I_{max} (mA)	**20**	20	40	20	100	10
Max.Current efficiency (%)	**99.96**	99.68	99.68	99.50	99.91	–
Load regulation (mV/mA)	**0.13**	–	0.19	–	0.09	0.01
Line regulation (mV/V)	**1.08**	–	2.67	–	0.744	0.52
PSR@1 MHz (dB)	**−30.8**	−23	−40	–	−15	–
Overshoot (mV)	**148**	88	122	95	200	15
Undershoot (mV)	**284**	35	127	90	297	15
Transient conditions	**100 μA–20 mA @1 ns**	100 μA–20 mA @5 ns	0–9 mA @0.35 ns	100 μA–20 mA @10 ns	10 μA–100 mA @100 ns	100 μA–20 mA @1 μs
1% Settling time (ns)	**841/734**	100/160	17/33	100/150	680	1100
FoM (ps)	**0.984**	15.75	12.86	17.52	5.63	–

[a]Simulation results.
[b]Measurement results.

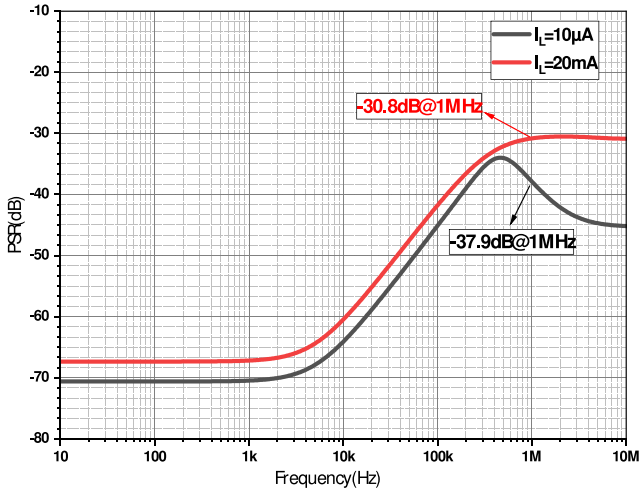


Fig. 16. Simulated PSR with C_L = 100 pF at minimum and maximum load current.

condition since the HPT compensates for the loss of gain compared to light load condition. With a 100 pF load capacitance, the stability requirements of the proposed LDO are fulfilled over a load current range of 10 μA to 20 mA. In addition, it can maintain stability over a wide range of load capacitances from 100 pF - 2 nF. The circuit structure is very simple and does not require complex frequency compensation to maintain stability. Thus, the proposed LDO is suitable for digital circuits in SoC applications.

CRediT authorship contribution statement

Yu Zhang: Design, Writing, Analysis, Reviewing and editing, Original draft preparation, Data curation. **Xiao Zhao:** Reviewing and editing. **Yongqing Wang:** Original draft preparation, Investigation. **Liyuan Dong:** Reviewing and editing.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence

our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled "A Fully Integrated Overshoot-Reduction Low-dropout Regulator Based on Hybrid NMOS/PMOS Power Transistors Technique for SoC Applications".

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We have approved the final version to be published. We agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

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